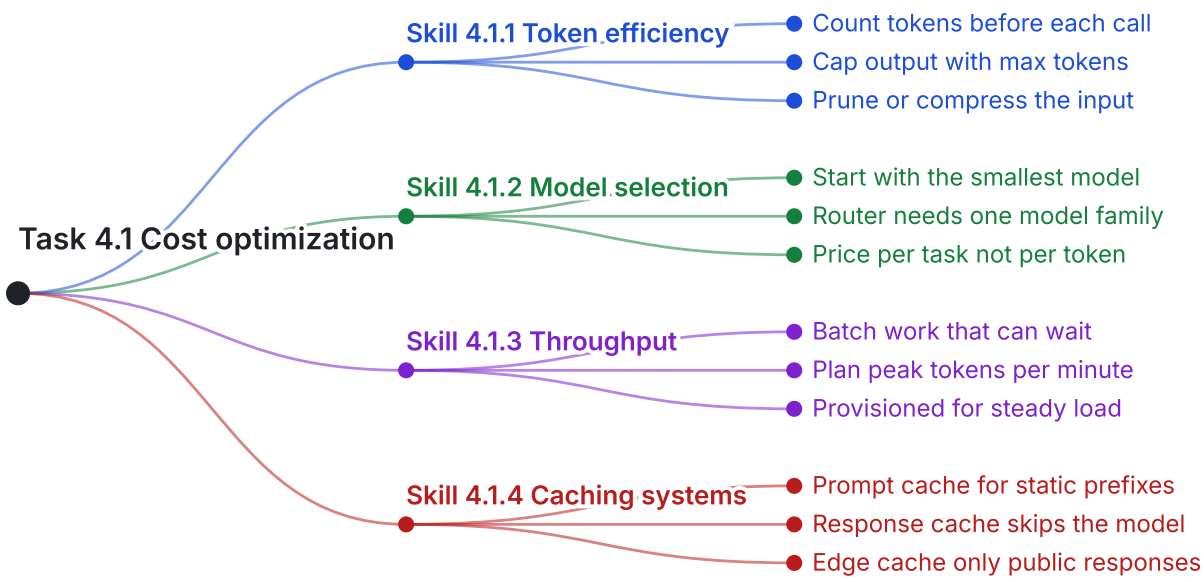
Memory sheet: Task 4.1 Implement cost optimization and resource efficiency strategies

Domain 4, weight 12 percent. Read this sheet, then answer the flash questions aloud.



Deciding rules

When the stem says	Think of	Not when
Nightly, overnight, backlog, or by morning	Batch inference with JSONL input in S3	A live user is waiting for each reply
Live chat, typing, or an agent waiting	Interactive call path with streaming for readers	The work can finish later in a batch job
The same long policy text repeats on every call	Prompt caching with a checkpoint after the static block	The whole answer repeats, so a response cache fits better
The same answer is wanted again for many users	Response cache, exact hash or semantic match	The answer depends on the user or the account
Steady, reserved, or guaranteed capacity	Provisioned Throughput for the steady load	The load spikes, since idle hours still bill
Spiky volume or unpredictable peaks	On-demand with quota planning and token budgets	A customized model needs dedicated capacity
Easy and hard queries mixed within one model family	Intelligent prompt routing with a fallback model	The models come from different families, so use a cascade
Large gigabyte volume from a private subnet through NAT	Compare the NAT path with an interface endpoint	The workload still needs internet access through NAT

Etch lines

1. Find the cost driver first, then pick the lever that touches it.
2. Read for the number that grows before you pick a network fix.
3. CountTokens gives the input count with no charge.
4. A high max tokens value reserves quota at once. Fewer requests fit each minute.
5. Cache reads do not count toward quota. Cache writes and input tokens still do.
6. Streaming changes first-word speed, not the token bill.

Flash questions

- Q: A live assistant resends the same long rules block every turn. Which feature cuts that input cost, and does the model still run? A: Prompt caching. The model still runs on each call.
- Q: A nightly job reads records from S3, and no person waits for results. Which invocation style fits? A: Batch inference with JSONL input and S3 output.
- Q: Which cache skips the model call entirely, and which one only reuses the input prefix? A: A response cache skips the call. Prompt caching only reuses the prefix.
- Q: A team sets max tokens very high to be safe. What does that do to requests per minute? A: It reserves quota at the start, so fewer requests fit.
- Q: Which API counts the input tokens of a request before the call, and does it carry a charge? A: CountTokens. It carries no charge.
- Q: Two candidate models come from different families, and the team wants per-request routing. Is the managed router the right tool? A: No. Use an application cascade instead.
- Q: A private workload sends large payloads through a NAT gateway. What should you compare before you switch to an endpoint? A: The monthly gigabytes and the hourly charges for each path.
- Q: A capacity plan counts only requests per minute. What risk does that miss? A: Long prompts with long outputs can exhaust the token quota first.

Source: Skill sections 4.1.1 to 4.1.4, Network cost in the request path, Versus pairs in this task, Read a long question, and Drill. AWS docs and the study guide sections above, no practice-test text.